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Short-term exposure to noise, fine particulate matter and nitrogen oxides on ambulatory blood pressure: A repeated-measure study

Li-Te Chang [a,1], Kai-Jen Chuang [b,c,1], Wei-Ting Yang [d], Ven-Shing Wang [d], Hsiao-Chi Chuang [e,f], Bo-Ying Bao [g], Chiu-Shong Liu [h], Ta-Yuan Chang [d,*]

[a] Department of Environmental Engineering and Science, Feng Chia University, Taichung, Taiwan
[b] Department of Public Health, School of Medicine, College of Medicine, Taipei Medical University, Taipei, Taiwan
[c] School of Public Health, College of Public Health and Nutrition, Taipei Medical University, Taipei, Taiwan
[d] Department of Occupational Safety and Health, College of Public Health, China Medical University, Taichung, Taiwan
[e] School of Respiratory Therapy, College of Medicine, Taipei Medical University, Taipei, Taiwan
[f] Division of Pulmonary Medicine, Department of Internal Medicine, Shuang Ho Hospital, Taipei Medical University, Taipei, Taiwan
[g] Department of Pharmacy, College of Pharmacy, China Medical University, Taichung, Taiwan
[h] Department of Family Medicine, China Medical University Hospital, Taichung, Taiwan

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ABSTRACT

Exposure to road traffic noise, fine particulate matter (aerodynamic diameter ≤ 2.5 μm; $PM_{2.5}$) and nitrogen oxides (NO_x) has been associated with transient changes in blood pressure, but whether an interaction exists remains unclear. This panel study investigated whether noise, $PM_{2.5}$ and NO_x exposure were independently associated with changes in 24-h ambulatory blood pressure. We recruited 33 males and 33 females aged 18–32 years as study subjects. Personal noise exposure and ambulatory blood pressure were monitored simultaneously in 2007. During the data collection periods, 24-h data on $PM_{2.5}$ and NO_x from five air-quality monitors within 6 km of participants' home addresses were used to estimate their individual exposures. Linear mixed-effects regression models were used to estimate single and combined effects on ambulatory blood pressure. Exposure to both noise and $PM_{2.5}$ was significantly associated with increased systolic blood pressure (SBP) and diastolic blood pressure (DBP) over 24 h; NO_x exposure was only significantly related to elevated DBP. Twenty-four-hour ambulatory blood pressure increased with the current noise exposure of 5 A-weighted decibels (dBA) (SBP 1.44 [95% confidence interval: 1.16, 1.71] mmHg and DBP 1.40 [1.18, 1.61] mmHg) and $PM_{2.5}$ exposure of 10-μg/m^3 (SBP 0.81 [0.19, 1.43] mmHg and DBP 0.63 [0.17, 1.10] mmHg), as well as the current NO_x exposure of 10-ppb (DBP 0.54 [0.12, 0.97] mmHg) after simultaneous adjustment. These findings suggest that exposure to noise and air pollutants may independently increase ambulatory blood pressure and the risk of cardiovascular diseases.

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1. Introduction

Road traffic is a major environmental pollutant that generates both annoying noise and contaminated air. Epidemiological studies have reported that exposure to road traffic noise, fine particulate matter (aerodynamic diameter ≤ 2.5 μm, $PM_{2.5}$) and nitrogen oxides (NO_x) was associated with transient blood pressure effects. Noise exposure above 55 A-weighted decibels (dBA) may increase systolic blood pressure (SBP) and diastolic blood pressure (DBP) in young and middle-aged adults (Chang et al., 2009, 2015). Short-term exposure to $PM_{2.5}$ was associated with increased SBP (Brook et al., 2011; Dvonch et al., 2009). Doubling NO_x exposure during one year preceding enrollment was associated with decreased SBP (Sorensen et al., 2012).

Residents who live along roadsides are always exposed to noise, $PM_{2.5}$ and NO_x simultaneously; they come from the same sources and make similar contributions to the development of cardiovascular diseases. However, their interactions with ambulatory blood pressure remain unclear. The possible biological mechanisms for the effects of $PM_{2.5}$ on blood pressure have been suggested to include local and systemic inflammation and

Abbreviations: DBA, A-weighted decibel; Leq, equivalent sound level; NO_x, nitrogen oxides; $PM_{2.5}$, particulate matter with aerodynamic diameter ≤ 2.5 μm; TWA, time-weighted average

* Correspondence to: Department of Occupational Safety and Health, College of Public Health, China Medical University, 91 Hsueh-Shih Road, Taichung 40402, Taiwan, R.O.C.. Fax: +886 4 22079225.
E-mail address: tychang@mail.cmu.edu.tw (T.-Y. Chang).

[1] Li-Te Chang and Kai-Jen Chuang contributed equally to this article.

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oxidative stress, autonomic imbalance and endothelial dysfunction (Brook and Rajagopalan, 2009; Brook et al., 2010; Mills and Snabre, 2009). Noise may activate the autonomic nervous and endocrine systems to produce the irregular endocrine and metabolic dysfunction that leads to hypertension (Babisch, 2011; Ising et al., 1999). Because $PM_{2.5}$ exposure and noise exposure may affect the same target system, the autonomic nervous system, it is worthwhile to investigate whether these agents have independent effects on blood pressure. Therefore, the purpose of this study was to explore whether noise, $PM_{2.5}$ and NO_x exposure independently affected adults' 24-h ambulatory blood pressure readings.

2. Materials and methods

2.1. Subjects

The participant recruitment and selection processes are described in detail elsewhere (Chang et al., 2009, 2012). Before the recruitment of study subjects, a commercial software (PASS, Kaysville, UT, USA) was used to calculate the minimum size of number for estimating the effect. Briefly, young adults were recruited through an on-campus advertisement at the China Medical University in 2007. A total of 69 students responded to the advertisement, but only 66 volunteers were included because three respondents reported having diagnosed hypertension on a questionnaire. No participants had any cardiovascular or pulmonary diseases. The major source of environmental noise came from road traffic; all study subjects lived at least 10 km from an airport.

In addition, a standardized questionnaire was used to collect individual information related to potential hypertension risk factors during the study period. These factors included age, height, weight, exercise habits, cigarette smoking, alcohol consumption, tea consumption, coffee consumption and a family history of hypertension. Cigarette smokers were defined as those who had smoked at least 3 days per week for more than 6 months. The same criterion was applied to define alcohol, tea and coffee drinkers. In addition, regular exercisers were defined as those who had participated in a sporting activity at least 3 times per week for 6 months or more (Chang et al., 2009, 2011, 2012). Each participant's height and weight were also gathered to calculate each individual's body mass index (BMI). During the study period, each subject was required to refrain from certain activities to prevent interference between the noise exposure measurements and the ambulatory blood pressure monitoring; prohibited activities were cigarette smoking, alcohol, tea and coffee consumption, drug use, regular exercise and the use of portable media players (Chang et al., 2009, 2012). This study was approved by the Institutional Review Board of the School of Public Health (China Medical University) before the study began, and informed consent was obtained from each participant.

2.2. Air pollutants monitoring

All participants' home addresses and the locations of 5 air-quality monitoring stations in Taichung City that had been set up around the study area by the Taiwan Environmental Protection Administration (Taiwan EPA) were identified using the geographic information system (GIS) software in ArcGIS 9.3.1 (Environmental Systems Research Institute Incorporated, Redlands, California, USA). The distances between individual home addresses and the 5 monitoring stations were calculated to have a range from 0.6 to 5.9 km, as shown in Fig. 1. Hourly monitoring data on $PM_{2.5}$ and NO_x levels were collected and assigned to each subject based on the closest station to his or her home address for the 24-h exposure measurements during the study period. This study was conducted on weekdays from Tuesday to Thursday from September to November.

2.3. Noise exposure measurements

Each participant's noise exposure was measured continuously using a personal noise dosimeter (Logging Noise Dose Meter Type 4443, Brüel & Kjær, Nærum, Denmark) that was set to record 5-min continuous equivalent sound levels (Leq) at an exchange rate of 3 dBA and to calculate the time-weighted-averages (TWA) of noise doses. The range of 50–120 dBA was chosen to measure all subjects' noise levels with 5-min readings over 24 h. Because noise levels were determined to be below the detected limits at night, a value of 35.4 dBA (the $\sqrt{2}/2$ of the detected noise level limit of 50 dBA) was used to replace the undetected values for this study (Hornung and Reed, 1990). To investigate the current and time-lagged effects of noise exposure, 5-min exposure measurements were summarized into 60-min and 120-min time-moving-average segments for further analysis.

2.4. Ambulatory blood pressure monitoring

A portable, non-invasive, automated monitoring and recording system (DynaPulse model 5000A, Pulse Metric, San Diego, California, USA) was used to monitor each subject's ambulatory SBP and DBP over 24 h. The ambulatory blood pressure measurements were automated continuously every 30 min during the daytime (08:00–23:00) and every 60 min at night (23:00–08:00) to avoid interrupting sleep quality. The DynaPulse system can measure an individual's arterial pulsation signals via a non-invasive cuff device that transfers the arterial waveform to the estimated ambulatory blood pressure. The SBP and DBP readings generated by this system were validated against the measurements using a traditional mercury sphygmomanometer (Brinton et al., 1998a, 1998b). All participants' ambulatory SBP and DBP readings during the daytime, at night and over 24 h were used to investigate the transient and sustained effects of exposure to noise, $PM_{2.5}$ and NO_x before and after simultaneous adjustment.

2.5. Statistical analysis

The Shapiro–Wilk test was used first to determine the normality of the continuous variables, i.e., age, BMI, and 24-h noise exposure, hourly $PM_{2.5}$ level and hourly NO_x level measurements. Since none of these variables were normally distributed among participants (all p Values < 0.001 for the Shapiro–Wilk test), a 10-based logarithm transformation was applied to fit the normal distribution before the regression models were employed. Accordingly, the non-parametric Spearman correlation method was used to calculate the coefficients between each exposure and ambulatory blood pressure during the daytime, at night and over 24 h.

Additionally, to explore the linear associations between ambulatory blood pressure and exposure to noise, $PM_{2.5}$ and NO_x, smoothing splines (following the GAMM procedure, mgcv-package, R version 2.13.2, R Foundation for Statistical Computing, Vienna, Austria) (Wood, 2004, 2006) were used in this study.

Next, linear mixed-effects regression models were applied to investigate the associations between exposure to noise, $PM_{2.5}$ and NO_x and ambulatory blood pressure by controlling for important participant confounding factors (Littell et al., 1996). These regression models were selected to address the auto-correlation problem between the repeated ambulatory blood pressure measurements and to increase statistical power by combining information across study subjects. Linear mixed-effects regression models have both

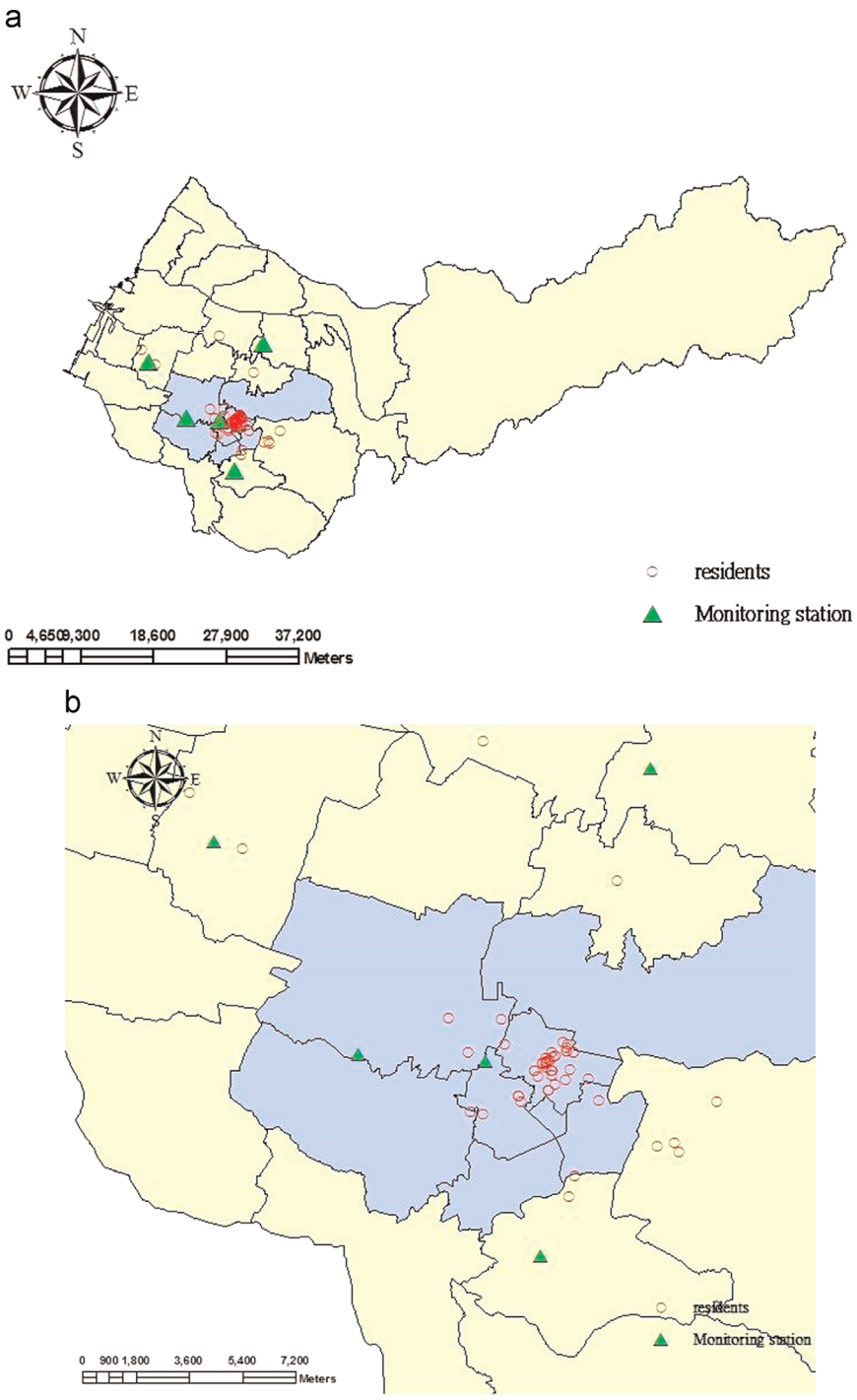


Fig. 1. All subjects' home addresses and the locations of 5 air-quality monitoring stations in Taichung City.

fixed-effect and random-effect covariance parameters. The fixed-effect variables were sex, age, BMI, cigarette smoking, regular exercise, measured levels of noise, $PM_{2.5}$ and NO_x exposure and family history of hypertension; sex, age, BMI, cigarette smoking, regular exercise and family history of hypertension are potential risk factors for hypertension that have been reported in previous studies (Chang et al., 2009, 2011; Jarup et al., 2008; Leon Bluhm et al., 2007). Individual subject was regarded as a random effect in this model. Influence and residual diagnostics were performed to ascertain that all assumptions of the mixed-effects regression models were met in the analyses.

Finally, the models were used to compare the within-period differences in the mean ambulatory SBP and DBP values and the average levels of noise, $PM_{2.5}$ and NO_x exposure during the daytime and at night. A compound symmetry covariance structure was chosen rather than the first-order autoregressive type because of the smaller Akaike's information criterion values for both ambulatory SBP and DBP (Cnaan et al., 1997). This study applied the MIXED procedure containing fixed and random effects in SAS version 9.1 (SAS Institute Incorporated, Cary, North Carolina, USA) to implement the linear mixed-effects regression models.

3. Results

Table 1 shows the demographic description of the study subjects. The gender proportions were equal, and no subjects reported

Table 1
Description of study subjects.

Characteristics	Subjects ($n=66$)		
	Mean (SD)	*N*	(%)
Age (years)	22.2 (2.2)		
Body mass index (kg/m^2)	23.0 (3.7)		
Gender, male		33	50.0
Current smoking, yes		6	9.1
Alcohol consumption, yes		0	0.0
Tea consumption, yes		33	50.0
Coffee consumption, yes		8	12.1
Regular exercise, yes		45	68.2
Family history of hypertension, yes		32	48.5

SD, standard deviation.

habitual alcohol consumption.

Table 2 presents the mean values and standard deviations for ambulatory blood pressure and noise and air pollutants levels among all subjects during the daytime, at night and over 24 h. Significantly higher mean ambulatory SBP and DBP values were found during the daytime compared with the measurements taken at night. The same significant results were observed in the average levels of noise, $PM_{2.5}$ and NO_x exposure.

The correlations between noise, $PM_{2.5}$ and NO_x exposure levels during different periods are shown in Appendix Table A1. Noise levels were significantly correlated with 24-h mean $PM_{2.5}$ and NO_x levels, with low correlation coefficients (rho was 0.08 and 0.09, respectively; both *p* Values <0.05). The mean $PM_{2.5}$ levels had moderate correlation with the mean NO_x levels during the daytime, at night and over 24 h (rho ranged from 0.25 to 0.34; all *p* Values <0.05). In addition, the mean noise, $PM_{2.5}$ and NO_x levels were significantly and positively correlated with the mean SBP and DBP values during the daytime and over 24 h (rho ranged from 0.08 to 0.27; all *p* Values <0.05), as shown in Appendix Table A2. At night, only noise and NO_x levels were significantly but weakly correlated with the mean ambulatory DBP (both correlation coefficients were 0.11 with *p* Values <0.05).

Smoothing splines were used to investigate the dose–response relationships between single exposure and ambulatory blood pressure. As shown in Fig. 2, levels of both noise and $PM_{2.5}$ exposure were significantly and positively associated with ambulatory SBP and DBP (both *p* Values <0.001 for noise; *p* Values were 0.013 and 0.003 for $PM_{2.5}$, respectively). However, exposure to NO_x was only significantly and positively related to increased ambulatory DBP in this study (*p* Values were 0.632 and 0.036 for SBP and DBP, respectively).

Table 3 shows the acute effects of single exposure to noise, $PM_{2.5}$ or NO_x on 24-h ambulatory blood pressure. Both ambulatory SBP and DBP significantly increased with a 5-dBA increment in the current, 60-min and 120-min time-lagged noise exposures. The greatest effects on ambulatory SBP and DBP were both observed after the 120-min time-lagged exposure. For $PM_{2.5}$, each 10-μg/m^3 increment was significantly associated with increased SBP and DBP for current, 60-min and 120-min time-lagged exposures. However, a 10-ppb increment in NO_x was only significantly related to increased DBP after the current and 1-h time-lagged exposures in this study. Similar patterns were identified only for noise exposure during the daytime (Appendix Table A3) but for both noise and NO_x exposure at night (Appendix Table A4).

Table 4 presents the associations between the combined exposure and 24-h ambulatory blood pressure for all subjects. Significant increases in SBP and DBP for the current, 60- and 120-min time-lagged noise and $PM_{2.5}$ exposures, as well as significantly elevated DBP after the current and 30-min time-lagged NO_x exposures were found in this study. Each 5-dBA increase in current noise exposure was significantly associated with increases of 1.44 (95% confidence interval: 1.16, 1.71) mmHg in SBP and 1.40 (1.18, 1.61) mmHg in DBP after adjusting for the 24-h mean $PM_{2.5}$ and NO_x levels. These effects on SBP (1.89 [1.57, 2.21] mmHg) and DBP (1.47 [1.23, 1.72] mmHg) were still observed to have the highest increase for the 120-min time-lagged noise exposure. For $PM_{2.5}$, each 10-μg/m^3 increment in current exposure was also significantly associated with increases of 0.81 (0.19, 1.43) mmHg in SBP and 0.63 (0.17, 1.10) mmHg in DBP after adjusting for 24-h mean noise and NO_x levels. The elevated effects on SBP (0.90 [0.13, 1.68] mmHg) and DBP (0.61 [0.06, 1.17] mmHg) of a 10-μg/m^3 increment in $PM_{2.5}$ persisted for the 120-min time-lagged exposure. However, each 10-ppb increment in NO_x exposure was only significantly associated with elevated DBP for the current (0.54 [0.12, 0.97] mmHg) and 60-min time-lagged (0.62 [0.15, 1.10] mmHg) exposures after simultaneous adjustment. Similar results were observed only for noise exposure during the daytime (Appendix Table A5) but for both noise and NO_x exposure at night (Appendix Table A6). In addition, both gender and BMI were significantly associated with elevated 24-h mean SBP and DBP, independent to noise or air pollutants levels (data not shown, all *p* Values <0.05).

4. Discussion

The present study demonstrated that short-term exposure to noise, $PM_{2.5}$ and NO_x may independently increase adults' 24-h ambulatory blood pressure. This was the first study to observe that acute noise exposure was associated with transient increases in ambulatory SBP and DBP after adjusting for $PM_{2.5}$ and NO_x levels. The results were consistent with the findings in two previous studies that did not control for air pollutants; one reported similar effects of SBP 1.43 (1.19, 1.67) mmHg and DBP 1.40 (1.20, 1.60) mmHg among young adults who were exposed to a 5-dBA noise increase (Chang et al., 2009), and the other found lower

Table 2
Ambulatory blood pressure, noise and air pollutants exposure during the daytime, at night and over 24 h.

	Daytime (08:00–23:00)		Nighttime (23:00–08:00)		24 h average	
	Mean (SD)	*N*[a]	Mean (SD)	N[a]	Mean (SD)	*N*[a]
Systolic blood pressure (mmHg)	123.8 (19.5)[b]	916	113.4 (18.2)	493	120.2 (19.7)	1409
Diastolic blood pressure (mmHg)	73.9 (13.8)[b]	916	64.8 (12.7)	493	70.7 (14.1)	1409
Noise (dBA)	61.5 (16.0)[c]	916	48.5 (14.2)	493	57.0 (16.6)	1409
$PM_{2.5}$ (μg/m^3)	34.1 (19.0)[c]	916	29.4 (20.2)	493	32.5 (19.6)	1409
NO_x (ppb)	26.1 (18.9)	916	22.9 (22.3)	493	25.0 (20.2)	1409

dBA, A-weighted decibel; NO_x, nitrogen oxides, $PM_{2.5}$, particulate matter with aerodynamic diameter ≤ 2.5 μm; and SD, standard deviation.

[a] Number of measurements.

[b] Differences ($p<0.05$) in ambulatory blood pressure were significant between daytime and nighttime (reference) after adjusting for noise levels, age, body mass index, cigarette smoking, regular exercise and family history of hypertension in the linear mixed-effects regression models.

[c] Differences ($p<0.05$) in exposure levels were significant between daytime and nighttime (reference) in the regression models.

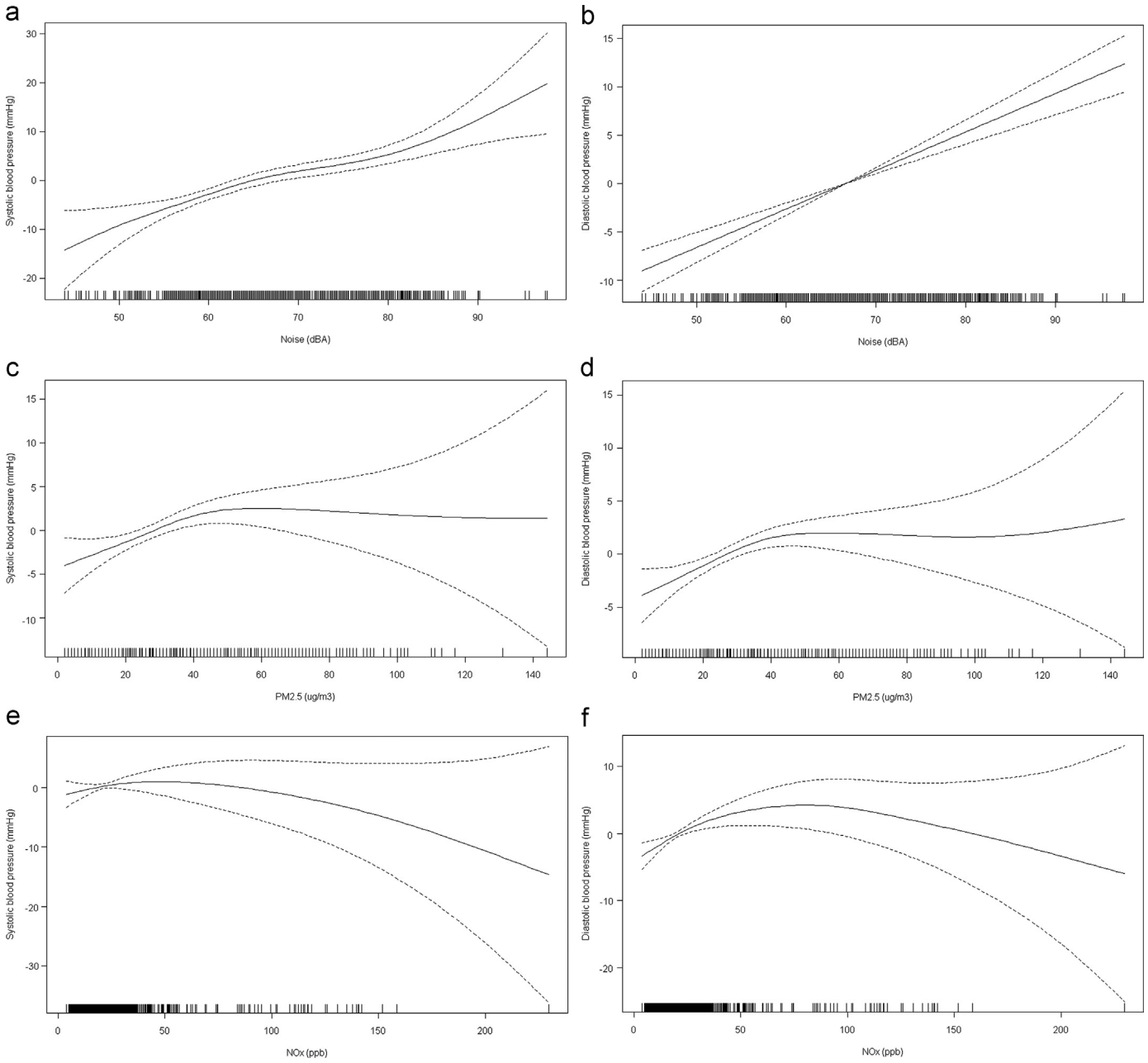


Fig. 2. Smoothing splines for the adjusted associations between ambulatory systolic blood pressure (SBP) and diastolic blood pressure (DBP) and exposure to noise, fine particulate matter (aerodynamic diameter ≤ 2.5 μm, $PM_{2.5}$) and nitrogen oxides (NO_x). All linear mixed-effects regression models were adjusted for age, body mass index, cigarette smoking, regular exercise and family history of hypertension.

increases of SBP 0.60 (0.30, 0.95) mmHg and DBP 0.80 (0.60, 1.05) mmHg with each 5-dBA noise exposure among normotensive workers aged 46 ± 4 years on their non-working days (Chang et al., 2015). The acute effects of noise exposure on ambulatory SBP and DBP persisted from the current to the 1-h and 2-h time-lagged exposures, indicating that transient increases in ambulatory blood pressure may lead to sustained elevated blood pressure and subsequent hypertension following repeated and prolonged exposure (Barregard et al., 2009; Chang et al., 2011; Leon Bluhm et al., 2007; van Kempen and Babisch, 2012).

Exposure to $PM_{2.5}$ also resulted in elevated effects on ambulatory SBP and DBP before and after adjusting for noise and NO_x levels in this study. These findings were consistent with the results from a population-based prospective cohort study that observed the long-term effects of $PM_{2.5}$ exposure on arterial SBP and DBP after controlling for noise exposure among adults aged 45–75 years with 14% diabetes mellitus (Fuks et al., 2011). However, only increased SBP was reported in two other previous studies; they did not take into account noise and NO_x exposure (Brook et al., 2011; Dvonch et al., 2009). These inconsistent results might be due to different approaches to exposure assessment and different strategies for measuring blood pressure (i.e., resting vs. ambulatory). Future studies should consider these factors to investigate the association between exposure to $PM_{2.5}$ and variations in blood pressure.

In addition, exposure to NO_x was found to increase 24-h ambulatory DBP after the current and 1-h time-lagged exposures. In contrast, significant decreases in SBP caused by doubled NO_x exposure during the 1- and 5-year periods preceding enrollment were reported in a population-based cohort study with subjects aged 50–64 years (Sorensen et al., 2012). Non-significant decreases in ambulatory SBP caused by the current, 1-h and 2-h time-lagged

Table 3
Associations between single exposure and 24-h ambulatory blood pressure among all subjects.

Model[a]	Increase in SBP (mmHg)	95% CI	Increase in DBP (mmHg)	95% CI
Noise (5-dBA increase)				
Current	1.43	1.15, 1.71	1.39	1.18, 1.61
60-min lag	1.76	1.46, 2.06	1.46	1.22, 1.69
120-min lag	1.88	1.56, 2.20	1.47	1.23, 1.72
$PM_{2.5}$ (10-μg/m^3 increase)				
Current	0.77	0.16, 1.37	0.68	0.24, 1.13
60-min lag	0.84	0.16, 1.53	0.71	0.21, 1.21
120-min lag	0.83	0.09, 1.58	0.67	0.15, 1.19
NO_x (10-ppb increase)				
Current	−0.11	−0.65, 0.44	0.52	0.11, 0.93
60-min lag	−0.10	−0.71, 0.52	0.59	0.13, 1.05
120-min lag	−0.22	−0.88, 0.44	0.47	−0.01, 0.95

CI, confidence interval; dBA, A-weighted decibel; DBP, diastolic blood pressure; NO_x, nitrogen oxides; $PM_{2.5}$, particulate matter with aerodynamic diameter ≤2.5 μm; and SBP, systolic blood pressure.

[a] Linear mixed-effects regression models were used with adjustments for sex, age, body mass index, cigarette smoking, regular exercise and family history of hypertension.

Table 4
Associations between combined exposure and 24-h ambulatory blood pressure among all subjects.

Model	Increase in SBP (mmHg)	95% CI	Increase in DBP (mmHg)	95% CI
Noise (5-dBA increase)[a]				
Current	1.44	1.16, 1.71	1.40	1.18, 1.61
60-min lag	1.76	1.47, 2.06	1.46	1.22, 1.69
120-min lag	1.89	1.57, 2.21	1.47	1.23, 1.72
$PM_{2.5}$ (10-μg/m^3 increase)[b]				
Current	0.81	0.19, 1.43	0.63	0.17, 1.10
60-min lag	0.90	0.19, 1.61	0.64	0.11, 1.17
120-min lag	0.90	0.13, 1.68	0.61	0.06, 1.17
NO_x (10-ppb increase)[c]				
Current	−0.12	−0.68, 0.44	0.54	0.12, 0.97
60-min lag	−0.12	−0.75, 0.52	0.62	0.15, 1.10
120-min lag	−0.26	−0.94, 0.42	0.49	−0.01, 0.99

CI, confidence interval; dBA, A-weighted decibel; DBP, diastolic blood pressure; NO_x, nitrogen oxides; $PM_{2.5}$, particulate matter with aerodynamic diameter ≤2.5 μm; and SBP, systolic blood pressure.

[a] The linear mixed-effects regression models were adjusted for 24-h average $PM_{2.5}$, 24-h average NO_x, sex, age, body mass index, cigarette smoking, regular exercise and family history of hypertension.

[b] The linear mixed-effects regression models were adjusted for 24-h average noise levels, 24-h average NO_x, sex, age, body mass index, cigarette smoking, regular exercise and family history of hypertension.

[c] The linear mixed-effects regression models were adjusted for 24-h average noise levels, 24-h average $PM_{2.5}$, sex, age, body mass index, cigarette smoking, regular exercise and family history of hypertension.

NO_x exposures were also observed in the present study. Differences in exposure assessment (stationary monitoring vs. modeling) and blood pressure measurements (ambulatory vs. resting) as well as in study designs (panel vs. cohort) might also have contributed to these inconsistent results.

The greater elevations of ambulatory SBP and DBP that were found with noise exposure after $PM_{2.5}$ and NO_x levels were adjusted for compared with the measurements that only considered a single exposure (both increases of 0.01 mmHg; 0.7%) indicated potential confounding effects of traffic-related air pollutants on blood pressure if these factors were not controlled for. In contrast, the increased effects of $PM_{2.5}$ exposure on ambulatory blood pressure were stronger for SBP (0.04 mmHg, 5.2%) but weaker for DBP (−0.05 mmHg, −7.4%) compared with the results from single-exposure models after adjusting for noise and NO_x levels. In addition, exposure to NO_x resulted in more elevated ambulatory DBP (0.02 mmHg, 3.8%) after adjusting for noise and $PM_{2.5}$ levels simultaneously. These results showed possible underestimated effects on ambulatory SBP when either noise or $PM_{2.5}$ exposure was not controlled for and on ambulatory DBP when noise and NO_x exposure were not adjusted for simultaneously. Therefore, future studies are recommended that investigate the effects of traffic-related noise or air pollutants on ambulatory blood pressure to consider this co-exposed interaction.

The current study also observed that gender and BMI were associated with increased 24-h ambulatory blood pressure. Male subjects and those with higher BMI had significantly higher means of SBP and DBP after adjusting for noise or air pollutants exposure, which were consistent with the findings in previous studies (Barregard et al., 2009; Chang et al., 2009; Jarup et al., 2008).

The strengths of this study included the use of repeated-measure analyses to explore the variations in ambulatory blood pressure caused by instant noise or air pollutants exposure. In addition, this study design explained the variance between and within individuals. The larger amount of time-series exposure and outcomes data solved the temporal problem between exposure to noise, $PM_{2.5}$ or NO_x and changes in ambulatory blood pressure.

Some limitations existed in the present study. First, monitoring data rather than personal sampling were used to represent individual exposure to $PM_{2.5}$ and NO_x. The use of individual noise-exposure measurements but station-based $PM_{2.5}$ and NO_x monitoring levels could have resulted in the greater effects on ambulatory SBP and DBP from noise exposure compared with the results for $PM_{2.5}$ or NO_x exposure. Accordingly, smaller effects on ambulatory SBP and DBP caused by air pollutants exposure were found in this study. However, exposure to noise, $PM_{2.5}$ and NO_x were significantly and positively associated with increased ambulatory blood pressure before and after simultaneous adjustment.

Secondly, the distance between environmental monitoring stations and dwellings (median: 2.2 km; 1st quartile: 2.0 km; 3rd quartile: 2.7 km) may have little exposure variations to subjects. This flaw restricted to the measurements of hourly air pollutants (i.e., $PM_{2.5}$ and NO_x), but did not affect the measurements of noise levels and ambulatory blood pressure since noise exposure and blood pressure were individually monitored during 24 h. In the current study, each subject was required to carry both the noise dosimeter and a monitoring device of ambulatory blood pressure in the research laboratory, and returned to take off them after 24 h. Accordingly, most subjects preferred to stay at home except sometimes having to study in school. Even subjects needed to study in school, it was not possible to exceed 6 h because they had to set up portable devices on their body. Therefore, we could assume that subjects spent a large proportion of time staying at home.

Thirdly, there were some imperfections in data collection. For the original design, 66 subjects were monitored hourly during 24 h to have a sum of 1584 data (i.e., 66*24=1584). But a total of 1409 values (1409/1584=89.0%) were collected and used to be analyzed here. The median of the missing proportion of measurements was 8.3% (1st quartile: 4.2%; 3rd quartile: 12.5%). Future studies are recommended to reduce the missing percentage.

Fourthly, the lack of sleep information (i.e., hours and quality of sleep) may overestimate acute effects of exposure to noise, $PM_{2.5}$ or NO_x on ambulatory blood pressure. The 1-h extension of sleep was found to be significantly associated with the decreased 24-h SBP and DBP among pre-hypertensive or hypertensive patients

aged 25–65 years (Haack et al., 2013). Sleep disturbance or poor quality of sleep may increase the nocturnal SBP and DBP in middle-aged pre-hypertensive or hypertensive subjects (Henskens et al., 2011; Sherwood et al., 2011). Future studies are suggested to collect this information for adjustment.

Finally, the small increases in ambulatory blood pressure induced by exposure to noise, $PM_{2.5}$ or NO_x among health adults cannot conclude to cause a hypertensive case clinically. Meanwhile, these results indicated that external environmental factors might aggravate the damage to susceptible subjects, such as patients with cardiovascular diseases. For instance, occupational noise exposure was found to raise higher ambulatory blood pressure among hypertensive adults compared with findings among normotensive ones (Chang et al., 2015). Exposures to $PM_{2.5}$ was associated with decreased heart rate variability, and the stronger effect of autonomic dysfunction caused by fine particle was found among people with ischemic heart disease, hypertension, and diabetes (Park et al., 2005). The authors believe that the development of cardiovascular diseases in cohort studies comes from the small increases in blood pressure after chronic exposure to these environmental factors.

5. Conclusions

Exposure to noise, $PM_{2.5}$ or NO_x may be independently associated with elevated ambulatory blood pressure in adults. Both noise and $PM_{2.5}$ exposure may cause increases in ambulatory SBP and DBP, but NO_x exposure only relates to elevated ambulatory DBP. In addition, the interactions of noise, $PM_{2.5}$ and NO_x exposure may contribute to the underestimation or overestimation of the effects of individual exposure on ambulatory SBP and DBP without simultaneous adjustment. Future epidemiological studies should consider the interactive effects of co-exposure to noise and air pollutants to investigate the association between individual exposure and the development of cardiovascular diseases.

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Conflicts of interest

None.

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Appendix A. Supplementary Information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.envres.2015.06.004.

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